Simplify before multiplying

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Calculate 2/3 x 9/10 using cancellation. Copy or complete the first step.

2. Calculate 2 1/2 x 3/4.

3. A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?

4. Miro says: Miro thinks multiplying always makes a number larger. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 2/3 x 9/10 using cancellation. Copy or complete the first step.
   Answer 3/5.

2. Calculate 2 1/2 x 3/4.
   Answer 1 7/8.

3. A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?
   Answer 2/5 m.

4. Miro says: Miro thinks multiplying always makes a number larger. Is this correct? Explain.
   Answer Multiplying by a proper fraction finds a part, so the result can be smaller than the starting number.

Simplify before multiplying

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ______________

1. Calculate 2/3 x 9/10 using cancellation.

2. Calculate 2 1/2 x 3/4.

3. A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?

4. Solve this independently and show every stage: Calculate 2/3 x 9/10 using cancellation.

5. Use a second method or representation for: Calculate 2 1/2 x 3/4.

6. Check this result using an inverse, estimate or known fact: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?

7. Prove or explain your reasoning: Calculate 4/7 of 21.

8. Identify and correct this misconception: Miro thinks multiplying always makes a number larger.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?

Teacher answers and checking prompts

1. Calculate 2/3 x 9/10 using cancellation.
   Answer 3/5.

2. Calculate 2 1/2 x 3/4.
   Answer 1 7/8.

3. A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?
   Answer 2/5 m.

4. Solve this independently and show every stage: Calculate 2/3 x 9/10 using cancellation.
   Answer 3/5.

5. Use a second method or representation for: Calculate 2 1/2 x 3/4.
   Answer Expected result: 1 7/8. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?
   Answer Expected result: 2/5 m. A valid check must be shown.

7. Prove or explain your reasoning: Calculate 4/7 of 21.
   Answer 12. The explanation must show why the method works.

8. Identify and correct this misconception: Miro thinks multiplying always makes a number larger.
   Answer Multiplying by a proper fraction finds a part, so the result can be smaller than the starting number.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Simplify before multiplying

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ________________

1. Prove or explain your reasoning: Calculate 4/7 of 21.

2. Prove the answer to this question, rather than only calculating it: Calculate 2/3 x 9/10 using cancellation.

3. Solve this in two different ways and compare them: Calculate 2 1/2 x 3/4.

4. Work backwards from the answer to reconstruct the question: 2/5 m.

5. Explain why this reasoning fails: Miro thinks multiplying always makes a number larger.

6. Create a new example that tests the same idea as: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate 4/7 of 21.
   Answer 12. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Calculate 2/3 x 9/10 using cancellation.
   Answer Expected result: 3/5. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 2 1/2 x 3/4.
   Answer Expected result: 1 7/8. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 2/5 m.
   Answer One valid reconstruction is: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro thinks multiplying always makes a number larger.
   Answer Multiplying by a proper fraction finds a part, so the result can be smaller than the starting number.

6. Create a new example that tests the same idea as: A ribbon is 3/5 m long. Maya uses 2/3 of it. How much ribbon does she use?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.